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Method for forming a contact plug by bottom-up metal growth

Abstract

A method is provided for forming a contact plug by bottom-up metal growth. In one step, a substrate is etched to form a contact hole that exposes a silicon-containing feature in the substrate. In one step, a silicide layer is formed on the silicon-containing feature. In one step, a metal seed layer is formed over the silicide layer. In one step, a metal contact layer is deposited over the metal seed layer to form the contact plug in the contact hole.

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Background/Summary

BACKGROUND

(1) The semiconductor integrated circuit (IC) industry has over the past decades experienced

tremendous advancements and is still experiencing vigorous development. With the dramatic advances in IC design, new generations of ICs have smaller and more complex structures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIG. 1 is a flow chart illustrating steps of a method for forming a contact plug by bottom-up metal growth in accordance with some embodiments.
- (3) FIGS. 2 through 17 are schematic views illustrating the steps of the method in accordance with some embodiments.
- (4) FIG. 18 is a schematic view illustrating a variation in accordance with some embodiments.
- (5) FIG. 19 is a schematic view illustrating another variation in accordance with some embodiments.

DETAILED DESCRIPTION

- (6) The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
- (7) Further, spatially relative terms, such as “on,” “above,” “over,” “downwardly,” “upwardly,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
- (8) With the advancement of semiconductor manufacturing technology, dimensions of semiconductor devices become smaller and smaller. Since the shrinkage of the device dimensions and/or device pitches may increase density of wires, a frontside portion of a wafer may have insufficient space to form all the wires therein. In order to address such a problem, some wires may be formed on a backside portion of the wafer, so the frontside portion of the wafer can have sufficient space for routing of the remaining wires. The wires that are selected to be formed on the backside portion of the wafer may be, for example but not limited to, power rails that may occupy larger spaces than other signal wires, other wires that are deemed suitable to be formed on the backside portion of the wafer by designers, or any combination thereof. In order to connect a wire that is formed on the backside portion of the wafer to a device that is formed on the frontside portion of the wafer, a metal contact plug may be formed in the backside portion of the wafer to electrically interconnect the wire and the device. To form the metal contact plug, a contact hole (e.g., a via hole or a trench, which is a long, narrow hole) would be formed in the backside portion of the wafer, and a metal layer is then deposited to fill the contact hole. In accordance with some

embodiments, the metal contact plug is connected to a non-metal feature of the device at the bottom of the contact hole, such as a source/drain region of a transistor (source/drain region(s) may refer to a source or a drain, individually or collectively dependent upon the context), and a growth rate of the metal layer on the non-metal feature may thus approximate to that on the sidewall of the contact hole, which is made of a dielectric material. Since the backside portion of the wafer is usually thick, the contact hole formed in the backside portion of the wafer usually has a large aspect ratio, which is a ratio of a depth to a width (e.g., a top width) of the contact hole. When the contact hole has a large aspect ratio and the growth rate of the metal layer on the sidewall of the contact hole approximates to that on the non-metal feature, which is exposed from the bottom of the contact hole, the contact hole may be sealed by the metal layer that is grown from the sidewall of the contact hole before the lower portion of the contact hole is completely filled, so voids and/or seams may be formed in the metal contact plug, resulting in high electrical resistance.

(9) FIG. 1 is a flow chart that cooperates with FIGS. 2-17 to illustrate steps of a method for forming a contact plug by bottom-up metal growth in accordance with some embodiments. The bottom-up metal growth can avoid formation of voids and/or seams in the contact plug, so as to achieve low electrical resistance for the contact plug. In the illustrative embodiment, the contact plug is formed on a backside portion of a substrate (e.g., a wafer substrate), but this disclosure is not limited in this respect. In accordance with some embodiments, the method may be used to form the contact plug on a frontside portion of the substrate.

(10) Referring to FIGS. 1 and 2, in step S1, a device wafer 1 is exemplarily provided to include a plurality of semiconductor devices on a frontside portion of a substrate 100. The substrate 100 may be a bulk semiconductor substrate or a semiconductor-on-insulator (SOI) substrate, which may be doped (e.g., with a p-type or an n-type dopant) or undoped. In some embodiments, an SOI substrate includes a layer of a semiconductor material formed on an insulator layer. The insulator layer may be a buried oxide (BOX) layer, a silicon oxide layer or any other suitable layer. The insulator layer may be provided on a suitable substrate, such as silicon, glass or the like. The substrate 100 may be made of a suitable semiconductor material, such as silicon or the like. In some embodiments, the substrate 100 is a silicon wafer; and in other embodiments, the substrate 100 is made of a compound semiconductor such as silicon carbide, gallium arsenide, indium arsenide, indium phosphide or other suitable materials. In still other embodiments, the substrate 100 is made of an alloy semiconductor such as GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, GaInAsP or other suitable materials.

(11) In some embodiments, the substrate 100 includes various p-type doped regions and/or n-type doped regions, such as p-type wells, n-type wells, p-type source/drain features and/or n-type source/drain features, formed by a suitable process such as ion implantation, thermal diffusion, a combination thereof, or the like. In some embodiments, the substrate 100 may include other functional elements such as resistors, capacitors, diodes, transistors, and/or the like. The transistors are, for example, field effect transistors (FETs), such as planar FETs and/or 3D FETs (e.g., FinFETs, GAAFETs). The substrate 100 may include lateral isolation features (e.g., shallow trench isolation (STI)) configured to separate various functional elements formed on and/or in the semiconductor substrate 100.

(12) In the illustrative embodiment, the semiconductor devices are gate-all-around field-effect transistors (GAAFETs), each including a gate electrode feature 110, multiple channel features 114 that are surrounded by the gate electrode feature 110, a gate dielectric 112 that is disposed between the gate electrode feature 110 and the channel features 114, a pair of source/drain features 116 that are disposed at opposite sides of the gate electrode feature 110 and that are connected to the channel features 114, and spacers 118 that are disposed between the gate electrode feature 110 and the source/drain features 116. The source/drain features 116 may refer to a source or a drain, individually or collectively dependent upon the context. In other embodiments, the semiconductor devices may include other types of circuit components, such as FinFETs, other suitable

components, or any combination thereof, and this disclosure is not limited in this respect.

(13) In accordance with some embodiments, the gate electrode feature **110** may include, for example, Cu, Ti, TiN, W, Al, Co, Ru, TiAlC, TaAlC, other suitable materials, or any combination thereof. In accordance with some embodiments, the channel features **114** may include, for example, Si, compound semiconductor, alloy semiconductor, other suitable materials, or any combination thereof. In accordance with some embodiments, the gate dielectric **112** may include, for example, a high-k material such as hafnium oxide, lanthanum oxide, etc., other suitable materials, or any combination thereof. In accordance with some embodiments, the source/drain features **116** are silicon-containing features that include silicon. In accordance with some embodiments, the source/drain features **116** are formed by, for example, epitaxial growth of silicon, other suitable techniques, or any combination thereof. In accordance with some embodiments, the spacers **118** may include, for example, silicon oxide, silicon nitride, oxygen-doped silicon nitride, carbon-doped silicon nitride, silicon carbide, other suitable low-k materials (e.g., having a dielectric constant smaller than 3.9), or any combination thereof. During the formation of the spacers **118**, an isolation layer **105** may also be formed on the substrate **100**, so as to enhance electrical isolation between the substrate **100** and the source/drain features **116**. In the illustrative embodiment, a first dielectric layer **122** is formed over the semiconductor devices, and several metal contacts **120** are formed in the first dielectric layer **122** to electrically connect the source/drain features **116** to other circuit elements. A second dielectric layer **126** is formed over the first dielectric layer **122**, a frontside interlayer dielectric (ILD) **130** is formed on the second dielectric layer **122**, and multiple metal wire layers **128** are formed in the frontside interlayer dielectric **130** and over the semiconductor devices in a frontside direction (upwards in FIG. 2). In the illustrative embodiment, one of the metal wire layers **128** is electrically connected to the gate electrode feature **110** of one of the semiconductor devices through a gate contact via **124** that is formed in the frontside interlayer dielectric **130**, the second dielectric layer **126** and the first dielectric layer **122**. In accordance with some embodiments, the first dielectric layer **122** may include, for example, silicon oxide, silicon nitride, silicon carbide, silicon oxynitride, silicon oxycarbide, a low-k material, other suitable materials, or any combination thereof. In accordance with some embodiments, the metal contacts **120** may include, for example, Co, W, Ru, Cu, Al, Mo, Ti, Ni, Au, Pt, Pd, other suitable materials, or any combination thereof. In accordance with some embodiments, the second dielectric layer **126** may include, for example, silicon oxide, silicon nitride, silicon carbide, silicon oxynitride, silicon oxycarbide, a low-k material, other suitable materials, or any combination thereof. In the illustrative embodiment, the first dielectric layer **122** is made of silicon dioxide or a low-k material for reducing a resistive-capacitive (RC) delay of the entire circuit, and the second dielectric layer **126** is made of silicon nitride to enhance electric isolation and to serve as an etch stop layer, but this disclosure is not limited to such. In accordance with some embodiments, the frontside interlayer dielectric **130** may include, for example, silicon oxide, silicon nitride, silicon oxynitride, silicon carbonitride, silicon carbide oxynitride, spin-on glass (SOG), phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), FSG, carbon doped silicon oxide (e.g., SiCOH), polyimide, other suitable materials, or any combination thereof. In accordance with some embodiments, the metal wire layers **128** and the gate contact via **124** may include, for example, Co, W, Ru, Cu, Al, Mo, Ti, Ni, Au, Pt, Pd, other suitable materials, or any combination thereof.

(14) Referring to FIGS. 1 and 3, in step S2, a carrier wafer **2** is bonded to the frontside portion of the device wafer **1** where the semiconductor devices are formed. The carrier wafer **2** includes a substrate **200**, and a bonding layer **20** formed on the substrate **200**. The substrate **200** may be a semiconductor substrate that is made of, for example, silicon, a compound semiconductor such as silicon carbide, gallium arsenide, indium arsenide, indium phosphide, an alloy semiconductor such as GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, GaInAsP, or other suitable materials. In the illustrative embodiment, the device wafer **1** and the carrier wafer **2** are connected together by oxide-oxide wafer bonding. The bonding layer **20** may be, for example, a silicon oxide (SiO_{sub}.x) layer. The

device wafer **1** is also provided with a bonding layer **160** (e.g., a silicon oxide layer) on top of the device wafer **1**, when viewed in the frontside direction. In accordance with some embodiments, the bonding layer **20** of the carrier wafer **2** may have a thickness greater than 200 angstroms to achieve good bonding.

(15) Referring to FIGS. **1** and **4**, in step **S3**, a backside portion of the substrate **100** is etched to form a contact hole **132**, wherein the contact hole **132** is disposed above the substrate **100** when viewed in a backside direction opposite to the frontside direction. The contact hole **132** exposes one of the source/drain features **116** (referred to as silicon-containing feature **116** hereinafter) that is disposed in the substrate **100**. In some embodiments, the substrate **100** may be the original semiconductor substrate as described above. In some embodiments, before step **S3**, the original semiconductor substrate may be removed from the backside of the device wafer **1**, and a dielectric material, such as SiO₂, may be deposited to form a new substrate **100**. In other words, in the subsequent backside process, the substrate **100** may be made of, for example, Si, SiO₂, other suitable materials, or any combination thereof. In accordance with some embodiments, the substrate **100** may be etched using, for example, wet etching, dry etching, other suitable techniques, or any combination thereof.

(16) Referring to FIGS. **1** and **5**, in step **S4**, a liner layer **134** is conformally formed on the backside portion of the substrate **100** and in the contact hole **132**. In accordance with some embodiments, the liner layer **134** may include, for example, SiN, SiCN, SiOCN, other suitable materials, or any combination thereof, and may be formed using, for example, atomic layer deposition (ALD), other suitable techniques, or any combination thereof. The liner layer **134** is used to prevent current leakage from the contact plug, which will be subsequently formed in the contact hole **132**, into the substrate **100**.

(17) Referring to FIGS. **1** and **6**, in step **S5**, when the backside direction is considered the upward direction, a portion of the liner layer **134** that is disposed at a bottom of the contact hole **132** and a portion of the liner layer **134** that is disposed over a top surface of the substrate **100** are etched, so that the silicon-containing feature **116** is revealed at the bottom of the contact hole **132**, while the remaining portion of the liner layer **134** is still attached to a sidewall of the contact hole **132**. In accordance with some embodiments, the liner layer **134** may be etched using an anisotropic dry etching technique, such as reactive-ion etching (RIE), ion beam etching (IBE), other suitable techniques, or any combination thereof.

(18) Referring to FIGS. **1** and **7** or FIGS. **1** and **8**, in step **S6**, a silicide layer **138** is formed over the silicon-containing feature **116**, so as to reduce contact resistance between the contact plug and the silicon-containing feature **116**, where FIG. **7** illustrates a case where the substrate **100** is made of silicon, and FIG. **8** illustrates a case where the substrate **100** is made of SiO₂. In detail, a preclean process is first performed to remove native oxide that is formed over the exposed silicon-containing feature **116**. In accordance with some embodiments, the preclean process may be performed using NF₃+NH₃ plasma to transform the native oxide into

(NH₄)₂SiF₆, and an annealing process to sublime the (NH₄)₂SiF₆.

After the preclean process, a metal film **136** is conformally deposited on the backside portion of the substrate **100** and in the contact hole **132**, and the metal film **136** reacts with the silicon-containing feature **116** to form the silicide layer **138** at an interface between the silicon-containing feature **116** and the metal film **136**. In accordance with some embodiments, the metal film **136** may include, Ti, Ni, Co, W, Pt, other suitable materials, or any combination thereof, so the silicide layer **138** may be, for example, Ti silicide, Ni silicide, Co silicide, W silicide, Pt silicide, other types of silicide, or any combination thereof. In accordance with some embodiments where the silicide layer **138** is made of Ti silicide, the silicide layer **138** may be formed using, for example, chemical vapor deposition (CVD) that employs TiCl₄ and hydrogen at a process temperature in a range from about 350° C. to about 500° C. In accordance with some other embodiments where the silicide layer **138** is made of Ti silicide, the silicide layer **138** may be formed using, for example, physical

vapor deposition (PVD) to conformally form a Ti layer on the backside portion of the substrate **100** and in the contact hole **132**, followed by an annealing process that may have a process temperature in a range from about 500° C. to about 600° C. to induce silicidation at the interface between the silicon-containing feature **116** and the Ti layer. However, this disclosure is not limited to specific methods or specific materials for forming the silicide layer **138**. Since the metal film **136** tends to be formed on silicon rather than on dielectric material, the metal film **136** is barely formed on the sidewall of the contact hole **132** in FIGS. **7** and **8** because of the liner layer **134**, which is made of a dielectric material. In FIG. **7**, the metal film **136** reacts with the silicon substrate **100**, so the silicide layer **138** is also formed at the interface between the metal film **136** and the substrate **100**. In FIG. **8**, the metal film **136** does not react with the SiO₂ substrate **100**, so the silicide layer **138** is formed only on the silicon-containing feature **116**, which is exposed at the bottom of the contact hole **132**.

(19) In accordance with some embodiments, after the deposition of the metal film **136**, a nitridation process is performed in step S6 to transform the metal film **136** into a metal nitride layer, so as to prevent underlayer oxidation. The nitridation process may include a plasma treatment that uses a gas including nitrogen, other suitable techniques, or any combination thereof. In accordance with some embodiments, the plasma treatment may be performed using, for example, N₂ plasma, N₂+H₂ plasma, other suitable techniques, or any combination thereof.

(20) Referring to FIGS. **1** and **9**, the metal nitride layer is etched to remove a portion of the metal nitride layer that is outside of the contact hole **132** and a portion of the metal nitride layer that is disposed on the sidewall of the contact hole **132**, while a portion of the metal nitride layer (referred to as protective layer **140** hereafter) that is disposed at the bottom of the contact hole **132** and over the silicide layer **138** remains. In accordance with some embodiments, the metal nitride layer may be etched using isotropic dry etching, other suitable techniques, or any combination thereof. The isotropic dry etching may use a chlorine-based etchant, such as BCl₃, but this disclosure is not limited in the respect. Because of the high aspect ratio of the contact hole **132**, the etching process can be controlled such that the protective layer **140** is not removed by the isotropic etching and remains over the silicide layer **138** to prevent the silicide layer **138** from oxidation. Since the silicide layer **138** and the protective layer **140** are both transformed from the metal film **136**, the silicide layer **138** and the protective layer **140** have the same metal element. In the case that the substrate **100** is a silicon substrate, a portion of the silicide layer **138** that is formed over the top surface of the substrate **100** (see FIG. **7**) may be removed after the etching of the metal nitride layer using, for example, wet etching, dry etching, other suitable techniques, or any combination thereof. In accordance with some embodiments, the silicide layer **138** formed at the bottom of the contact hole **132** may have a thickness in a range from about 3 nm to about 6 nm, and the protective layer **140** may have a thickness in a range from about 1 nm to about 2 nm, but this disclosure is not limited in this respect. An excessively thick protective layer **140** (e.g., greater than 2 nm in thickness) may result in undesired large electrical resistance between the subsequently formed contact plug and the silicon-containing feature **116**, and an excessively thin protective layer **140** (e.g., smaller than 1 nm in thickness) may not effectively prevent underlayer oxidation.

(21) Referring to FIGS. **1** and **10** through **12**, in step S7, a metal seed layer **142** is formed over the silicide layer **138** and the protective layer **140**. In FIG. **10**, a metal film (also denoted by the reference numeral **142**) is conformally deposited over the substrate **100** and in the contact hole **132**. In accordance with some embodiments, the metal film **142** may include, for example, W, Mo, Ru, Co, other suitable materials with low electrical resistivity, or any combination thereof, and may be formed using, for example, directional PVD, other suitable techniques, or any combination thereof, so that the metal film **142** is thick at the bottom of the contact hole **132** and is thin on the sidewall of the contact hole **132**. In accordance with some embodiments, a portion of the metal film **142** at the bottom of the contact hole **132** may have a thickness in a range from about 5 nm to about 8 nm. In FIG. **11**, a portion of the metal film **142** that is outside of the contact hole **132** and a portion of

the metal film **142** that is on the sidewall of the contact hole **132** is transformed into a metal oxide layer **144**. In accordance with some embodiments, the metal oxide layer **144** may be formed by performing a plasma treatment (e.g., **02** plasma) that uses a gas including oxygen on the metal film **142**, other suitable techniques, or any combination thereof. In accordance with some embodiments, the plasma treatment is performed with zero bias to make the plasma treatment non-directional, so that, because of the high aspect ratio of the contact hole **132**, a portion of the metal film **142** that is disposed at the bottom of the contact hole **132** will not be oxidized. In FIG. **12**, the metal oxide layer **144** is removed, and only the portion of the metal film **142** that is disposed at the bottom of the contact hole **132** remains on the metal nitride layer **140**, serving as the metal seed layer **142**. In accordance with some embodiments, the metal oxide layer **144** may be removed using, for example, isotropic dry etching, other suitable techniques, or any combination thereof. The isotropic dry etching may use, for example, a chlorine-based etchant, but this disclosure is not limited in this respect.

(22) Referring to FIGS. **1** and **13**, in step **S8**, a metal contact layer **143** is deposited over the metal seed layer **142**, so as to form the contact plug in the contact hole **132** (see FIG. **12**) in a manner of bottom-up growth. In accordance with some embodiments, the metal contact layer **143** may include, for example, W, Mo, Ru, Co, other suitable materials, or any combination thereof, and may be formed using, for example, CVD, ALD, other suitable techniques, or any combination thereof. By using the techniques such as CVD or ALD with a chlorine-base precursor, the metal thus deposited tends to be formed only on metals (e.g., the metal seed layer **142**), and is barely formed on dielectrics (e.g., the liner layer **134**), so the metal contact layer **143** is formed in a manner of bottom-up growth. In general, the bottom-up metal growth prevents formation of voids or seams in the metal contact layer **143**, so the resultant metal contact plug will have better electrical resistance. It is noted that the metal contact layer **143** and the metal seed layer **142** may be made of either the same material or different materials, and this disclosure is not limited in this respect. After the contact hole **132** is filled or almost filled with the metal contact layer **143**, an additional metal layer **146** is deposited on the metal contact layer **143** and the substrate **100** for the subsequent planarization process. In accordance with some embodiments, the metal layer **146** may be deposited using, for example, PVD, other suitable techniques, or any combination thereof. In FIG. **14**, a chemical-mechanical planarization (CMP) process is performed to planarize and expose the top surface of the substrate **100**, so as to form the contact plug **144** with a flat top surface, where the liner layer **134** is disposed between a lateral surface of the contact plug **144** and the substrate **100**.

(23) Referring to FIGS. **15** through **17**, a plurality of metal wire layers are exemplarily formed above the semiconductor devices when viewed in the backside direction. In FIG. **15**, a first metal wire layer **148A** is formed to be in contact with the contact plug **144**. In FIG. **16**, a second metal wire layer **148B** is formed to be electrically connected to the first metal wire layer **148A** through a first contact via **150A**. In FIG. **17**, a third metal wire layer **148C** is formed to be electrically connected to the second metal wire layer **148B** through a second contact via **150B**. In accordance with some embodiments, the metal wire layers **148A**, **148B**, **148C** and the contact vias **150A**, **150B** may include, for example, Co, W, Ru, Cu, Al, Mo, Ti, Ni, Au, Pt, Pd, other suitable materials, or any combination thereof, and are formed in a backside interlayer dielectric **151**. In accordance with some embodiments, the backside interlayer dielectric **151** may include, for example, silicon oxide, silicon nitride, silicon oxynitride, silicon carbonitride, silicon carbide oxynitride, SOG, PSG, BPSG, FSG, carbon doped silicon oxide (e.g., SiCOH), polyimide, other suitable materials, or any combination thereof. In accordance with some embodiments, the metal wire layers **148A**, **148B**, **148C** form power rails for transmission of electrical power to the semiconductor devices that are formed at the frontside of the substrate **100**, where the first metal wire layer **148A** is electrically connected to the silicon-containing feature **116**, which is the source/drain feature of the GAAFET in the illustrative embodiment.

(24) In the illustrative embodiment, the contact hole **132** (see FIGS. **4** through **12**) and the resultant contact plug **144** (see FIGS. **14** through **17**) has an aspect ratio (e.g., a height-to-width ratio) in a range from about 3 to about 8. If the aspect ratio is excessively small (e.g., smaller than 3), the portion of the metal nitride layer that is disposed at the bottom of the contact hole **132** may be removed during the etching of the metal nitride layer, and the portion of the metal film **142** that is disposed at the bottom of the contact hole **132** may be oxidized during the oxidation process. In such a condition, the protective layer **140** and/or the metal seed layer **142** may not be successfully formed, and the bottom-up metal growth of the contact plug **144** may not be realized. If the aspect ratio is excessively large (e.g., greater than 8), the reactants for forming the silicide layer **138**, the protective layer **140** and/or the metal seed layer **142** may not reach the bottom of the contact hole **132**, resulting in a low manufacturing yield.

(25) In accordance with some embodiments, when the backside direction is considered the upward direction, a top opening of the contact hole **132** and a top portion of the contact plug **144** may have a width in a range from about 10 nm to about 50 nm. For the top opening of the contact hole **132** and the top portion of the contact plug **144**, an excessively large width (e.g., greater than 50 nm) may increase difficulty in overlay control, and an excessively small width (e.g., smaller than 10 nm) may lead to difficulty in formation of the required films/layers at the bottom of the contact hole **132**. In accordance with some embodiments, a bottom of the contact hole **132** and a bottom portion of the contact plug **144** may have a width in a range from about 8 nm to about 45 nm. For the bottom of the contact hole **132** and the bottom portion of the contact plug **144**, an excessively large width (e.g., greater than 45 nm) may increase difficulty in overlay control, and an excessively small width (e.g., smaller than 8 nm) may lead to insufficient formation of the silicide layer **138** at the bottom of the contact hole **132**. In accordance with some embodiments, a height of the contact hole **132** and a height of the contact plug **144** may be determined based on the desired aspect ratio and the desired width for the top portion of the contact plug **144**, and usually fall within a range from about 10 nm to about 80 nm.

(26) FIG. **18** illustrates a first variation of a circuit structure of the abovementioned embodiments. In FIG. **18**, the device wafer **1** and the carrier wafer **2** are connected together by hybrid bonding. In such a scenario, the device wafer **1** has a frontside surface that is formed by a top metal layer **128A** and a top dielectric layer **130A**, the carrier wafer **2** has a frontside surface that is formed by a top metal layer **24** and a top dielectric layer **22**, and the top metal layer **128A** of the device wafer **1** is connected to the top metal layer **24** of the carrier wafer **2**. In accordance with some embodiments, the top metal layers **128A**, **24** of the device wafer **1** and the carrier wafer **2** may include, for example, Cu, other suitable materials, or any combination thereof. In accordance with some embodiments, the top dielectric layers **130A**, **22** of the device wafer **1** and the carrier wafer **2** may include, for example, SiON, other suitable materials, or any combination thereof. In accordance with some embodiments, the top dielectric layer **22** of the carrier wafer **2** may have a thickness greater than 200 angstroms to achieve good bonding. Since the top metal layers **128A**, **24** of the device wafer **1** and the carrier wafer **2** have good thermal conductivity, the hybrid bonding may be advantageous in terms of heat dissipation.

(27) FIG. **19** illustrates a second variation of a circuit structure of the abovementioned embodiments, which differs from the first variation in that the carrier wafer **2** has a passive component **26** formed therein. In the illustrative embodiment, the passive component **26** has a metal-insulator-metal (MIM) structure, which may be used as a memory component. The MIM structure includes two metal electrodes, and an insulator feature sandwiched between the metal electrodes. In accordance with some embodiments, the metal electrodes may include, for example, TiN, TaN, WN, other suitable materials, or any combination thereof, and the insulator feature may include, for example, a high-k material (e.g., having a dielectric constant greater than 3.9), HfO₂, HfSiO, HfON, HfZrO, HfAlO, other suitable materials, or any combination thereof.

(28) In summary, the contact plug **144** that is electrically connected to the silicon-containing

feature **116** is formed by forming the silicide layer **138** and the metal seed layer **142** at the bottom of the contact hole **132**, so as to induce bottom-up metal growth to prevent formation of voids and/or seams in the contact plug **144**, and the contact plug **144** may thus have good electric resistance. In accordance with some embodiments, the protective layer **140**, which is a metal nitride layer in the illustrative embodiments, is formed over the silicide layer **138** to prevent oxidation of the silicide layer **138** during the fabrication processes.

(29) In accordance with some embodiments, a method for forming a contact plug by bottom-up metal growth is provided. In one step, a substrate is etched to form a contact hole that exposes a silicon-containing feature disposed in the substrate. In one step, a silicide layer is formed over the silicon-containing feature. In one step, a metal seed layer is formed over the silicide layer. In one step, a metal contact layer is deposited over the metal seed layer to form the contact plug in the contact hole.

(30) In accordance with some embodiments, between the step of forming the silicide layer and the step of forming the metal seed layer, a metal nitride layer is formed over the silicide layer. The metal seed layer is formed over the metal nitride layer.

(31) In accordance with some embodiments, the step of forming the silicide layer includes depositing a metal film on the silicon-containing feature, and making the metal film react with the silicon-containing feature to form the silicide layer at an interface between the metal film and the silicon-containing feature. The metal nitride layer is formed by performing a nitridation process to transform the metal film into the metal nitride layer.

(32) In accordance with some embodiments, the nitridation process includes a step of performing a plasma treatment that uses a gas including nitrogen to transform the metal film into the metal nitride layer.

(33) In accordance with some embodiments, the step of forming the metal seed layer includes the following actions. In one action, a metal film is conformally deposited over the substrate and in the contact hole. In one action, a portion of the metal film that is outside of the contact hole and a portion of the metal film that is on a sidewall of the contact hole are transformed into a metal oxide layer. In one action, the metal oxide layer is removed.

(34) In accordance with some embodiments, the metal oxide layer is formed by performing a plasma treatment that uses a gas including oxygen on the metal film.

(35) In accordance with some embodiments, the plasma treatment is performed with zero bias.

(36) In accordance with some embodiments, the metal film is formed by directional physical vapor deposition (PVD).

(37) In accordance with some embodiments, the contact hole has an aspect ratio in a range from 3 to 8.

(38) In accordance with some embodiments, the metal contact layer is deposited using a chlorine-based precursor.

(39) In accordance with some embodiments, the substrate is formed with a metal wire layer above the silicon-containing feature when a first direction is considered an upward direction, and the contact hole is formed in the substrate above the silicon-containing feature when a second direction that is opposite to the first direction is considered the upward direction.

(40) In accordance with some embodiments, a circuit structure is provided to include a silicon-containing feature disposed in a substrate, a metal contact plug disposed in the substrate and electrically connected to the silicon-containing feature, and a silicide layer disposed between the silicon-containing feature and the metal contact plug.

(41) In accordance with some embodiments, the circuit structure further includes a metal nitride layer is disposed between the silicide layer and the metal contact plug.

(42) In accordance with some embodiments, the metal nitride layer and the silicide layer include a same metal element.

(43) In accordance with some embodiments, the circuit structure further includes a metal seed layer

that is disposed between the metal nitride layer and the metal contact plug.

(44) In accordance with some embodiments, the metal contact plug has an aspect ratio in a range from 3 to 8.

(45) In accordance with some embodiments, the circuit structure further includes a metal wire layer disposed over the silicon-containing feature in a first direction. The metal contact plug is disposed above the silicon-containing feature when a second direction opposite to the first direction is considered an upward direction.

(46) In accordance with some embodiments, the circuit structure further includes a dielectric liner layer disposed between a lateral surface of the metal contact plug and the substrate.

(47) In accordance with some embodiments, a method for forming a contact plug by bottom-up metal growth is provided. In one step, a substrate in which a silicon-containing feature is disposed is etched to form a contact hole that exposes a silicon-containing feature at a bottom of the contact hole. In one step, a silicide layer is formed at the bottom of the contact hole, wherein the silicide layer is in contact with the silicon-containing feature. In one step, a metal seed layer is formed at the bottom of the contact hole and over the silicide layer. In one step, chemical vapor deposition (CVD) is used to deposit a metal contact layer over the metal seed layer, so as to fill the contact hole and form the contact plug in the contact hole.

(48) In accordance with some embodiments, between the step of forming the silicide layer and the step of forming the metal seed layer, a metal nitride layer is formed at the bottom of the contact hole and over the silicide layer. The metal seed layer is formed over and in contact with the metal nitride layer.

(49) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method for forming a contact plug by bottom-up metal growth, comprising steps of: etching a substrate to form a contact hole that exposes a silicon-containing feature disposed in the substrate; after the contact hole is formed, transforming a portion of the silicon-containing feature that is exposed from the contact hole into a silicide layer; forming a metal seed layer over the silicide layer; and depositing a metal contact layer over the metal seed layer to form the contact plug in the contact hole.
2. The method according to claim 1, further comprising, between the step of forming the silicide layer and the step of forming the metal seed layer, a step of forming a metal nitride layer over the silicide layer; wherein the metal seed layer is formed over the metal nitride layer.
3. The method according to claim 2 wherein the step of forming the silicide layer includes: depositing a metal film on the silicon-containing feature; and making the metal film react with the silicon-containing feature to form the silicide layer at an interface between the metal film and the silicon-containing feature; and wherein the metal nitride layer is formed by performing a nitridation process to transform the metal film into the metal nitride layer.
4. The method according to claim 3, wherein the nitridation process includes a step of performing a plasma treatment that uses a gas including nitrogen to transform the metal film into the metal nitride layer.
5. The method according to claim 1, wherein the step of forming the metal seed layer includes:

conformally depositing a metal film over the substrate and in the contact hole; transforming a portion of the metal film that is outside of the contact hole and a portion of the metal film that is on a sidewall of the contact hole into a metal oxide layer; and removing the metal oxide layer.

6. The method according to claim 5, wherein the metal oxide layer is formed by performing a plasma treatment that uses a gas including oxygen on the metal film.

7. The method according to claim 5, wherein the metal film is formed by directional physical vapor deposition (PVD).

8. The method according to claim 5, wherein the contact hole has an aspect ratio in a range from 3 to 8.

9. The method according to claim 1, wherein the metal contact layer is deposited using a chlorine-based precursor.

10. The method according to claim 1, wherein the substrate is formed with a metal wire layer above the silicon-containing feature when a first direction is considered an upward direction, and the contact hole is formed in the substrate above the silicon-containing feature when a second direction that is opposite to the first direction is considered the upward direction.

11. A method for forming a contact plug by bottom-up metal growth, comprising steps of: etching a substrate in which a silicon-containing feature is disposed to form a contact hole that exposes a silicon-containing feature at a bottom of the contact hole; forming a silicide layer at the bottom of the contact hole, wherein the silicide layer is in contact with the silicon-containing feature; forming a metal nitride layer at the bottom of the contact hole and over the silicide layer, wherein the metal seed layer is formed over and in contact with the metal nitride layer; forming a metal seed layer over and in contact with the metal nitride layer; and depositing a metal contact layer over the metal seed layer, so as to fill the contact hole and form the contact plug in the contact hole, wherein the step of forming the silicide layer includes: conformally depositing a metal film on the substrate and in the contact hole, thereby forming the silicide layer at an interface between the silicon-containing feature and the metal film, and wherein the step of forming the metal nitride layer includes: transforming the metal film into a metal nitride film by a plasma treatment that uses a gas including nitrogen; and removing a portion of the metal nitride film that is outside of the contact hole and a portion of the metal nitride film that is disposed on a sidewall of the contact hole, thereby forming the metal nitride layer at the bottom of the contact hole and over the silicide layer.

12. The method according to claim 11, comprising, before the step of forming the silicide layer, a step of removing native oxide from the silicon-containing feature that is exposed in the contact hole.

13. The method according to claim 12, wherein the step of removing native oxide includes: transforming the native oxide into a salt; and sublimating the salt.

14. A method for forming a contact plug by bottom-up metal growth, comprising steps of: forming a semiconductor device on a frontside portion of a semiconductor substrate, where the semiconductor device includes a silicon-containing feature; etching a backside portion of the semiconductor substrate to expose a part of the silicon-containing feature from a backside of the semiconductor substrate; forming a silicide layer on the part of the silicon-containing feature; and forming a metal contact plug over the silicide layer with bottom-up growth.

15. The method according to claim 14, further comprising: forming a protective metal nitride layer over the silicide layer before forming the metal contact plug.

16. The method according to claim 15, further comprising: forming a metal seed layer over the protective metal nitride layer before forming the metal contact plug.

17. The method according to claim 14, wherein the silicide layer is formed after the step of etching the backside portion of the semiconductor substrate.

18. The method according to claim 14, wherein the step of forming the silicide layer includes: conformally depositing a metal film on the substrate and the part of the silicon-containing feature, thereby forming the silicide layer at an interface between the part of the silicon-containing feature

and the metal film, and wherein the step of etching the backside portion of the semiconductor substrate forms a contact hole that exposes the part of the silicon-containing feature, and the method further comprises steps of: transforming the metal film into a metal nitride film; and removing a portion of the metal nitride film that is outside of the contact hole and a portion of the metal nitride film that is disposed on a sidewall of the contact hole, thereby forming a metal nitride layer at the bottom of the contact hole and over the silicide layer.

19. The method according to claim 11, wherein the silicide layer is formed after the step of etching the substrate.

20. The method according to claim 1, wherein the step of transforming the portion of the silicon-containing feature into the silicide layer includes: conformally depositing a metal film on the substrate and the portion of the silicon-containing feature, thereby forming the silicide layer at an interface between the portion of the silicon-containing feature and the metal film, and wherein the method further comprises steps of: transforming the metal film into a metal nitride film; and removing a portion of the metal nitride film that is outside of the contact hole and a portion of the metal nitride film that is disposed on a sidewall of the contact hole, thereby forming a metal nitride layer at a bottom of the contact hole and over the silicide layer.
